BOUNDARY ELEMENT MODELING OF SOLID OXIDE MEMBRANE PROCESS

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Keywords: electrochemistry, mathematical modeling, boundary element method, magnesium, solid oxide membrane

Abstract

Mathematical modeling is a useful tool for design of electrochemical systems such as the magnesium solid-oxide membrane process (Mg-SOM). In this study, the Boundary Element Method (BEM) was particularly suited to the 3-dimensional modeling of the Mg-SOM process. BEM modeling allows one to change the geometry without remeshing, a useful feature for parametric optimization. After benchmarks demonstrated the accuracy of the program, the resulting model calculated electric current density for both single YSZ membrane and three YSZ membrane geometries of this process.

Introduction

The Solid Oxide Membrane (SOM) process is an energy efficient process for the extraction of metals directly from their oxides [1]. The magnesium solid oxide membrane process (Mg-SOM) offers a cost-efficient and clean alternative to existing technology for primary magnesium extraction. The Mg-SOM process utilizes a tubular yttria-stabilized zirconia membrane containing liquid metal as an anode at a temperature of 1150˚C. A molten ionic flux contains dissolved magnesium oxide and releases oxygen ions that are removed through the solid oxide membrane and oxidized at the tin anode. Magnesium vapor evolves at the cathode and is carried by inert gas (argon) to a condenser chamber [2, 3].

Proof of concept for this process was previously demonstrated using a 100g scale reactor and the viability of a larger-scale reactor is now being investigated [2, 3]. Mathematical modeling enables further understanding of the Mg-SOM system [4]. The Boundary Element Method (BEM) is particularly suited to the three-dimensional modeling of the Mg-SOM process. BEM modeling allows one to change the geometry without remeshing, a useful feature for parametric optimization [5]. A model is being developed to calculate the electric current density and heat generation rates for both single tube and three tube geometries of the Mg-SOM setup [6]. The model will enable the optimization of electrode placement, first to minimize total resistance of the system, and second to maximize current density uniformity at the anode. This study focuses on solving the equations for macroscopic current in geometries that contain a single zirconia tube or three zirconia tubes.

Methodology

Assumptions

Charge transfer effects were neglected because at the high temperatures utilized in the Mg-SOM process (1150°C), these effects are negligible in comparison to electrolyte resistance. Sufficient flux stirring ensures that mass transfer is not limiting. Diffusion through the thick membrane is the most limiting. Table I shows the values used to create the model's single and triple tube geometries.

	Single Tube	Triple Tube
Crucible Length	2.125”	2.125”
Crucible Diameter	1.575”	4.276”
Membrane Diameter	0.756”	0.756”
Membrane Length	1.63”	1.63”
Bubbling Tube Diameter	0.25”	0.25”
Bubbling Tube Length	1.25”	1.25”
Flux Conductivity	383 S/m	383 S/m
YSZ membrane Conductivity	10 S/m	10 S/m
Anode Surface Area	24.9 cm^2 (3.85 in^2)	74.7 cm^2 (11.55 in^2)

Table I: Dimensions Used in the Single and Triple Tube Models

Boundary Conditions

Dirichlet boundary conditions were applied to the cathode surfaces contacting the molten flux. These include the stainless steel crucible wall as well as the stainless steel bubbling tubes. Robin boundary conditions were applied to the anode surfaces contacting the molten flux. The anode surfaces include the exterior surfaces of the yttria-stabilized zirconia membrane. The top of the solid oxide membrane(s), bubbling tubes, and crucibles had natural boundary conditions applied to them, i.e. zero current density. Additionally, the bottom of the bubbling tubes were presumed to not be in contact with molten flux because of the constantly exiting vapor from the end of the tubes, and natural boundary conditions applied there as well.

Dirichlet Boundary Condition : $\Phi = A$ where A = constant

Robin Boundary Condition : $\nabla \Phi \cdot n = A + B\,\Phi$: where A = constant and B = constant

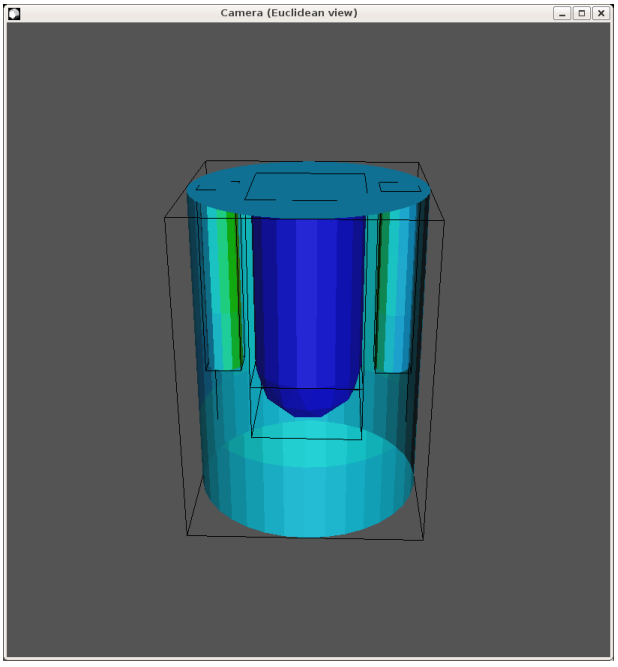


Figure 1: Mg-SOM 3-D Model Geometry: Single Tube

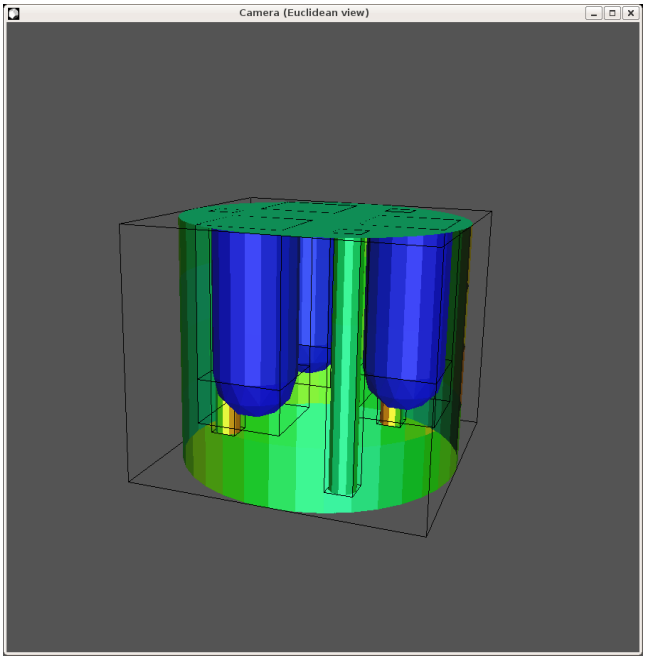


Figure 2: Mg-SOM 3-D Model Geometry: Triple Tube

Code was written to generate the 3-D geometries of both the single and triple membrane Mg-SOM cells. Figures 1 and 2 show screen shots of the two geometries. Figure 1 shows the single tube geometry and Figure 2 shows the triple tube geometry. Note that the 3-D models here include only the surfaces that are in direct contact with the molten flux.

Lead Wire Resistance

Lead wire resistance was determined by taking potentiodynamic scans of the individual components of the setup over a voltage range of 0 to 0.7 V and the corresponding resistance was calculated using Ohm's law. An average of the values shown in the plots from Figure 3 are shown in Table 2. For use in modeling, we summed the average resistance values for the Lead Wires, Stainless Cathode, and Molybdenum Tube (1150°C) (0.081 ohm). Based on these values, the lead resistance for the triple tube experiment was estimated to be 0.075 ohm.

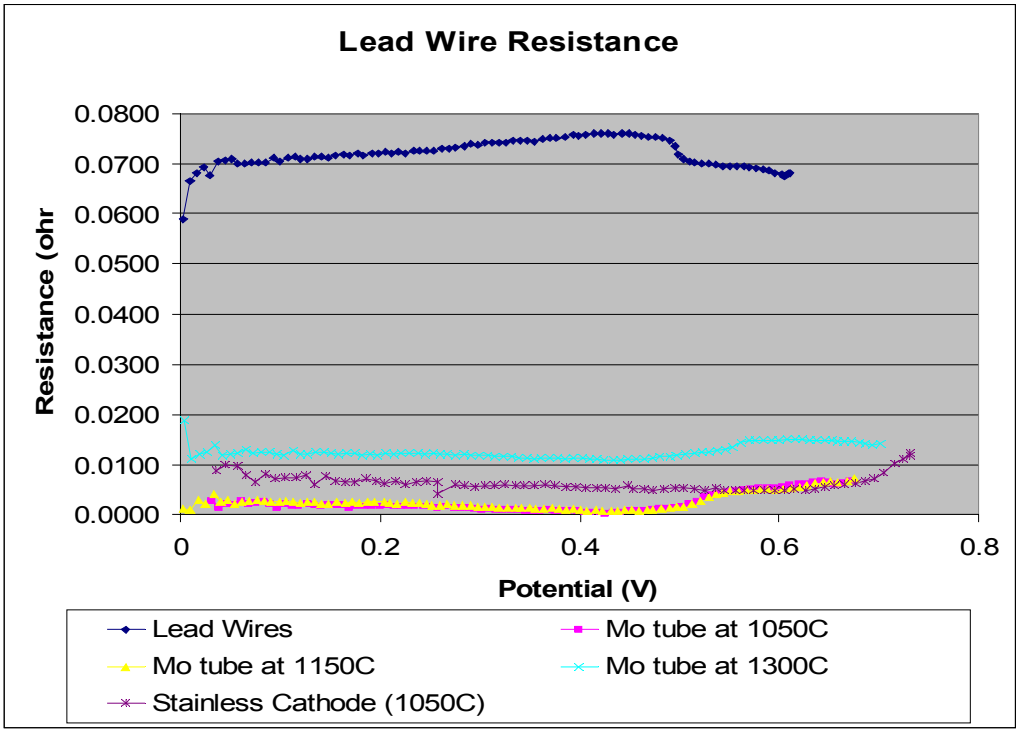


Figure 3: Plot of Resistance of Components of Experimental Setup.

	Average Resistance (ohms)	Standard Deviation
Lead Wires (connect potentiostat to anode/cathode)	0.0719	±0.0030
Stainless Steel Cathode	0.0064	±0.0017
Molybdenum Tube (1050°C)	0.0022	±0.0019
Molybdenum Tube (1150°C)	0.0027	±0.0017
Molybdenum Tube (1300°C)	0.0126	±0.0014

Table 2: Lead Wire Resistance Averages

Results

Simulated Potentiodynamic Scans for Single and Triple Tube Models

Calculations of the average current density using the BEM model fall within a realistic range. At an applied potential of 5 V the average anodic current density was calculated to be 0.9 A/cm^2 in both the single and triple tube simulations. This calculation takes into account a lead wire resistance value of 0.081 ohm and calculated flux resistance of 0.220 ohm. The 0.9 A/cm^2 calculated current density value is similar to the experimental value of 1 A/cm^2, validating the model and its assumptions. Further refinement of the model will allow us to predict the values more accurately. Figure 4 shows an experimental Mg-SOM potentiodynamic scan with modeling results from the single tube geometry. The modeled values were normalized to the 15.2 cm^2 anode area utilized in the experiment. By inspection, the experimental values for current response are somewhat lower than those calculated using the boundary element method. Figure 5 shows a plot of applied potential versus anode current values generated using the model for the triple tube geometry. The triple tube geometry plot used 74.7 cm^2 anode area.

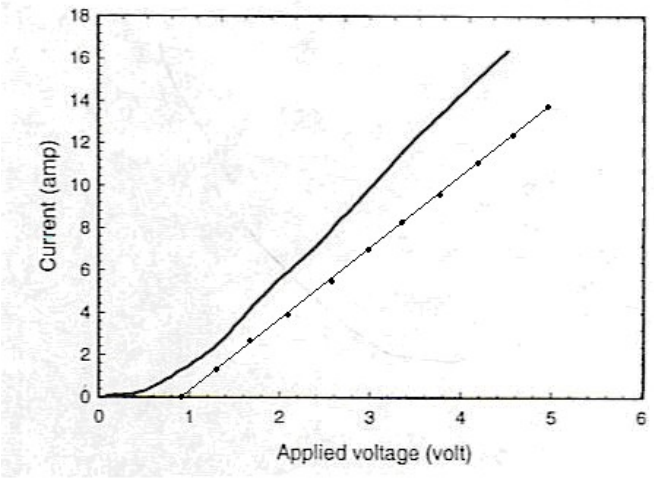


Figure 4. Potentiodynamic Scan from Mg-SOM experiment with single-tube modeling results.

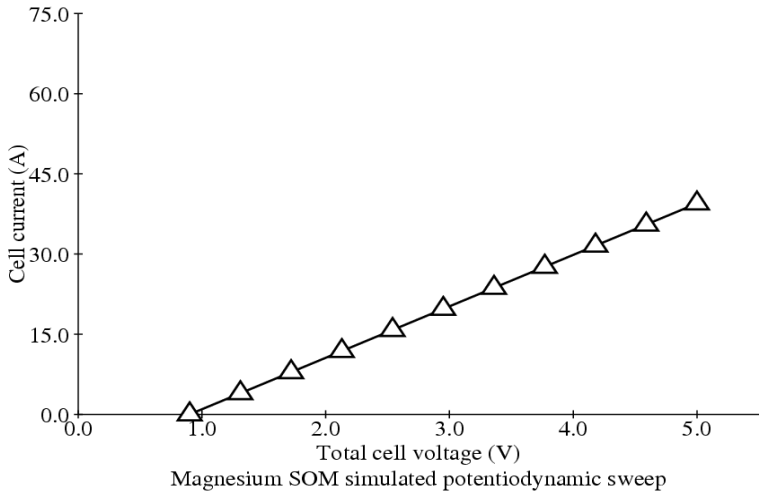


Figure 5: Calculated Values for Anode Current using Triple Tube BEM Model

Adjustment of Design Based on Simulation Results

The placement of bubbling tubes with respect to the YSZ membrane was identified as a factor affecting the current density distribution. Minimizing variation in current density distribution across the YSZ membrane is a factor in extending the life of the YSZ membrane. Variations in this distribution lead to uneven heating in the ceramic. Simulations were created that varied the depth of immersion of the bubbling tube from 1 inch to 2 inches (incremented by ¼ inch). The resulting current density distributions were compared. Figure 6 plots the depth of immersion versus the difference between absolute values of the maximum and minimum current density values calculated by the simulation. These results indicate that a more narrow distribution can be achieved by lowering the bubbling tube to within a half inch of the bottom of the 2.25 inch deep stainless steel crucible.

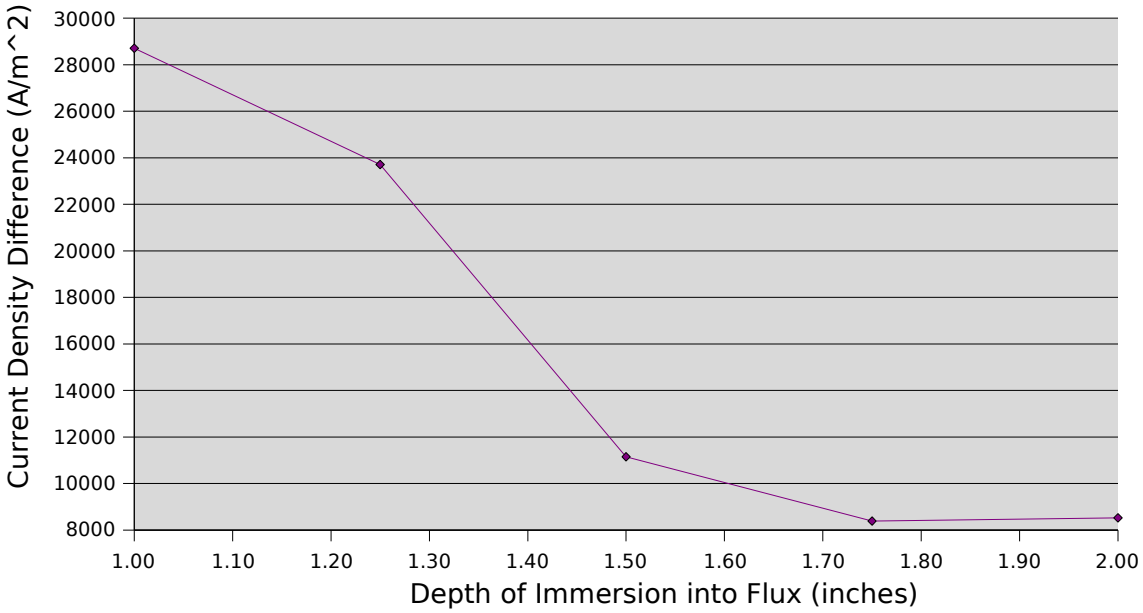


Figure 6: Current Density Difference vs. Depth of Bubbling Tube Immersion

Figures 7 and 8 show pictures of the single tube model at the minimum and maximum depths of bubbling tube immersion, respectively. This exercise illustrates the utility of the BEM model of the Mg-SOM process to the design of the experimental SOM reactor. A similar study of the effects of bubbling tube immersion experimentally would have taken upwards of 2 months to complete. While the model may not be perfectly accurate with respect to the experimental values, its does exhibit great usefulness in the examination of various design parameters.

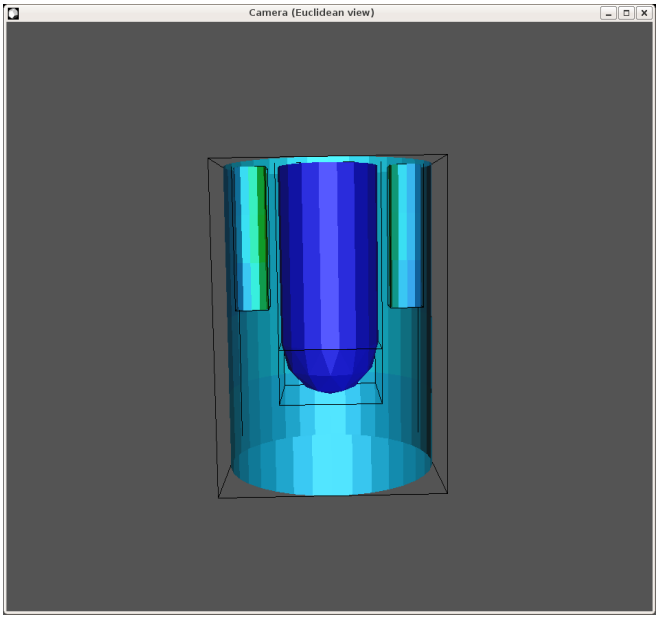


Figure 7: Bubbling Tube Immersion = 1 inch.

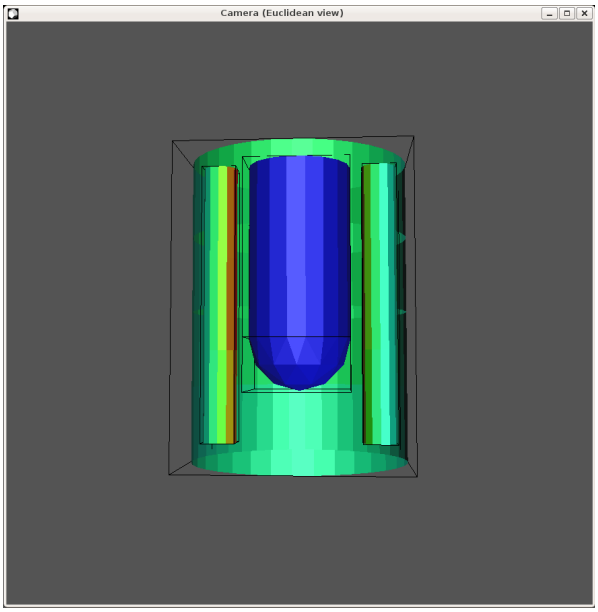


Figure 8: Bubbling Tube Immersion = 2 inches.

Future Work

The BEM model can now provide a platform for further investigation into the Mg-SOM process. Further work on this project will include the optimization of electrode placement, first to minimize total resistance of the system, and second to maximize current density uniformity at the anode. Calculations of the heat generation will also be incorporated. Additional SOM experiments will be made to test the simulation results of the triple tube model tube. A 39-tube simulation will be designed to investigate feasibility of designs for a pilot-scale reactor.

Acknowledgments

The authors thank Safe Hydrogen LLC and the Department of Energy (DOE) for funding this research with Award No. DE-FC36-04GO14011 and Contract No. C407200-AN8.

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